U18. Conjugate directions and linear conjugate gradient

E21 Numerical example *ch6, p. 10 (solved exercise 3)*

Nonzero vectors $\{p_0, \dots, p_k\}$ are A-conjugate if $p_i^\mathsf{T} A p_j = 0$ for all $i \neq j$. Let $A = \begin{pmatrix} 2 & 1 \\ 1 & 1.5 \end{pmatrix}$, $p_0 = (1,1)^\mathsf{T}$ and $p_1 = (1,-2)^\mathsf{T}$.

a) Decide whether p_0 and p_1 are A-conjugate.

b) Replace p_1 by $\tilde{p}_1 = p_1 - \dfrac{p_1^\mathsf{T} A p_0}{p_0^\mathsf{T} A p_0}\, p_0$ (one Gram–Schmidt step in the inner product $\langle u, v\rangle_A = u^\mathsf{T} A v$) and verify that $\tilde{p}_1 = (\frac{15}{11}, -\frac{18}{11})^\mathsf{T}$ is A-conjugate to p_0.

(a) Computing $p_0^\mathsf{T} A p_1$:

$$Ap_1 = \begin{pmatrix} 2 & 1 \\ 1 & 3/2 \end{pmatrix} \begin{pmatrix} 1 \\ -2 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 - 3 \end{pmatrix} = \begin{pmatrix} 0 \\ -2 \end{pmatrix}, \qquad p_0^\mathsf{T} A p_1 = (1,1) \begin{pmatrix} 0 \\ -2 \end{pmatrix} = -2.$$

$\therefore$ p_0 and p_1 are *not* A-conjugate.

(b)

- **Affirmation:** $p_0^\mathsf{T} A \tilde{p}_1 = 0$.

$$\tilde{p}_1 = p_1 - \frac{p_1^\mathsf{T} A p_0}{p_0^\mathsf{T} A p_0}\, p_0, \qquad A\tilde{p}_1 = Ap_1 - \frac{p_1^\mathsf{T} A p_0}{p_0^\mathsf{T} A p_0}\, Ap_0,$$

$$\begin{aligned} p_0^\mathsf{T} A \tilde{p}_1 &= p_0^\mathsf{T} A p_1 - \frac{p_1^\mathsf{T} A p_0}{p_0^\mathsf{T} A p_0}\, p_0^\mathsf{T} A p_0 \\ &= p_0^\mathsf{T} A p_1 - p_1^\mathsf{T} A p_0. \end{aligned}$$

  Since $A = A^\mathsf{T}$, $p_0^\mathsf{T} A p_1 = p_1^\mathsf{T} A p_0$, so $p_0^\mathsf{T} A \tilde{p}_1 = 0$.
  $\therefore$ p_0 and $\tilde{p}_1$ are A-conjugate.

- Computing $\tilde{p}_1$:

$$p_1^\mathsf{T} A p_0 = p_0^\mathsf{T} A p_1 = -2, \qquad Ap_0 = \begin{pmatrix} 2 & 1 \\ 1 & 3/2 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ 5/2 \end{pmatrix},$$

$$p_0^\mathsf{T} A p_0 = (1,1) \begin{pmatrix} 3 \\ 5/2 \end{pmatrix} = \frac{11}{2}, \qquad \frac{p_1^\mathsf{T} A p_0}{p_0^\mathsf{T} A p_0} = \frac{-2}{11/2} = -\frac{4}{11}.$$

$$\begin{aligned} \tilde{p}_1 &= p_1 - \frac{p_1^\mathsf{T} A p_0}{p_0^\mathsf{T} A p_0}\, p_0 = (1,-2) - \Big(-\frac{4}{11}\Big)(1,1) \\ &= \Big(1 + \frac{4}{11}, -2 + \frac{4}{11}\Big) = \Big(\frac{15}{11}, -\frac{18}{11}\Big). \end{aligned}$$

P17 Practice *added*

Why conjugacy matters. Let $\varphi(x) = \frac{1}{2}x^\mathsf{T}Ax - b^\mathsf{T}x$ with $A = \begin{pmatrix} 4 & 1 \\ 1 & 3 \end{pmatrix}$, $b = (1,2)^\mathsf{T}$, $x^* = A^{-1}b = (\frac{1}{11}, \frac{7}{11})^\mathsf{T}$ and $\varphi(x^*) = -\frac{15}{22}$. In every case below, $x_{k+1} = x_k + \alpha_k p_k$ with the exact step $\alpha_k = -\dfrac{(Ax_k - b)^\mathsf{T}p_k}{p_k^\mathsf{T}Ap_k}$, from $x_0 = (0,0)^\mathsf{T}$.

a) Cyclic coordinate directions $p_0 = e_1$, $p_1 = e_2$, $p_2 = e_1$, $p_3 = e_2$: compute $x_1, \dots, x_4$ and the gap $\varphi(x_k) - \varphi(x^*)$. Show that x^* is not reached in finitely many steps.

b) Directions $p_0 = e_1$ and $p_1 = e_2 - \dfrac{e_2^\mathsf{T}Ae_1}{e_1^\mathsf{T}Ae_1}e_1 = (-\frac{1}{4}, 1)^\mathsf{T}$: verify $p_0^\mathsf{T}Ap_1 = 0$ and that $x_2 = x^*$.

c) Steepest descent, $p_k = -(Ax_k - b)$: compute $x_1, \dots, x_4$ and the gap, and verify $p_{k+1} \perp p_k$.

d) Explain the difference between a) and b) in terms of the expansion $x^* - x_0 = \sum_i \sigma_i p_i$.

Computations are done on the computer. Check **P17.py** [Check computation with table V18, they match]

(a)

```
[ITEM A] Cyclic coordinate directions
step= 0, x0=[0.000 0.000],x1=[0.250 0.000], p=[1 0], alpha=  0.250,gap= 5.568e-01, df(x0)@p=-1.00, df(x1)=[
0.000 -1.750]
step= 1, x0=[0.250 0.000],x1=[0.250 0.583], p=[0 1], alpha=  0.583,gap= 4.640e-02, df(x0)@p=-1.75,
df(x1)=[0.583 0.000]
step= 2, x0=[0.250 0.583],x1=[0.104 0.583], p=[1 0], alpha= -0.146,gap= 3.867e-03, df(x0)@p= 0.58,
df(x1)=[-0.000 -0.146]
step= 3, x0=[0.104 0.583],x1=[0.104 0.632], p=[0 1], alpha=  0.049,gap= 3.222e-04, df(x0)@p=-0.15,
df(x1)=[0.049 0.000]
step= 4, x0=[0.104 0.632],x1=[0.092 0.632], p=[1 0], alpha= -0.012,gap= 2.685e-05, df(x0)@p= 0.05,
df(x1)=[-0.000 -0.012]
step= 5, x0=[0.092 0.632],x1=[0.092 0.636], p=[0 1], alpha=  0.004,gap= 2.238e-06, df(x0)@p=-0.01,
df(x1)=[0.004 0.000]
step= 6, x0=[0.092 0.636],x1=[0.091 0.636], p=[1 0], alpha= -0.001,gap= 1.865e-07, df(x0)@p= 0.00, df(x1)=[
0.000 -0.001]
step= 7, x0=[0.091 0.636],x1=[0.091 0.636], p=[0 1], alpha=  0.000,gap= 1.554e-08, df(x0)@p=-0.00,
df(x1)=[0.000 0.000]
```

(gap) $\varphi(x_k) - \varphi(x^*) \to 0$.

(direction) `df(x0)@p` is $\nabla\varphi(x_k)^\mathsf{T}p_k$. We have $\nabla\varphi(x_k)^\mathsf{T}p_k > 0$ for $k = 2, 4$: p_k is not a descent direction.

Why $\|\nabla\varphi(x_k)\| \to 0$? $\nabla\varphi(x_k)$ is `df(x1)` on the console.

(b)

```
[ITEM B] Conjugated directions
step= 0, x0=[0.000 0.000],x1=[0.250 0.000], p=[1.000 0.000], alpha=  0.250,gap= 5.568e-01, df(x0)@p=-1.00,
df(x1)=[ 0.000 -1.750]
step= 1, x0=[0.250 0.000],x1=[0.091 0.636], p=[-0.250  1.000], alpha=  0.636,gap= 0.000e+00, df(x0)@p=-1.75,
df(x1)=[0.000 0.000]
xn=[0.091 0.636], ||xn-x*||=0.0
```

$$x_n = x^*, \qquad n = 2.$$

(c)

```
[ITEM C] Steepest descent directions
step= 4, x0=[0.000 0.000],x1=[0.250 0.500], p=p1=[1.000 2.000], p1@p0=0.0 , alpha=  0.250,gap= 5.682e-02,
df(x0)@p1=-5.00, df(x1)=[ 0.500 -0.250]
step= 3, x0=[0.250 0.500],x1=[0.083 0.583], p=p1=[-0.500  0.250], p1@p0=0.0 , alpha=  0.333,gap= 4.735e-03,
df(x0)@p1=-0.31, df(x1)=[-0.083 -0.167]
step= 2, x0=[0.083 0.583],x1=[0.104 0.625], p=p1=[0.083 0.167], p1@p0=-2.9802322387695312e-08 , alpha=
0.250,gap= 3.946e-04, df(x0)@p1=-0.03, df(x1)=[ 0.042 -0.021]
step= 1, x0=[0.104 0.625],x1=[0.090 0.632], p=p1=[-0.042  0.021], p1@p0=4.7031790018081665e-08 , alpha=
0.333,gap= 3.284e-05, df(x0)@p1=-0.00, df(x1)=[-0.007 -0.014]
step= 0, x0=[0.090 0.632],x1=[0.092 0.635], p=p1=[0.007 0.014], p1@p0=-3.3178366720676422e-09 , alpha=
0.250,gap= 2.736e-06, df(x0)@p1=-0.00, df(x1)=[ 0.003 -0.002]
xn=[0.092 0.635], ||xn-x*||=0.0014551187884905695
```

We observe $p_k^\mathsf{T} p_{k-1} \approx 0$ (`p1@p0`).

Since $p_k = -\nabla f(x_k)$, we confirm

$$\nabla f(x_k)^\mathsf{T} p_k = \texttt{df(x0)@p1} < 0,$$

so p_k is a descent direction.

(d) For (b), $\sigma_i = \alpha_i$; that is why (b) reaches x^* after at most n steps. That is not the case for (a): p_i and p_j are not A-conjugate.

T49 PROOF *ch6, Theorem 2.1, pp. 2–3*

Setting. Let $\varphi(x) = \frac{1}{2}x^\mathsf{T}Ax - b^\mathsf{T}x$ with $A \in \mathbb{R}^{n\times n}$ symmetric positive definite, $x^* = A^{-1}b$, and residual $r_k = Ax_k - b = \nabla\varphi(x_k)$.
Problem. Let $p_0, \ldots, p_{n-1}$ be nonzero and A-conjugate ($p_i^\mathsf{T} A p_j = 0$ for $i \neq j$), $x_0 \in \mathbb{R}^n$ arbitrary, and $x_{k+1} = x_k + \alpha_k p_k$ with $\alpha_k = -\dfrac{r_k^\mathsf{T} p_k}{p_k^\mathsf{T} A p_k}$.

a) Prove that $p_0, \ldots, p_{n-1}$ are linearly independent.

b) Prove that α_k is the exact minimizer of $\alpha \mapsto \varphi(x_k + \alpha p_k)$.

c) Prove that $r_k^\mathsf{T} p_i = 0$ for all $i < k$, and that $x_n = x^*$.

Note. The notes use $\{p_i\}$ as a basis without proving linear independence (part a), which is what makes the expansion $x^* - x_0 = \sum_i \sigma_i p_i$ possible.

(a) Let's suppose that $\{p_0, \ldots, p_{n-1}\}$ are linearly dependent

$$\Rightarrow\ \exists\, k \in \{0, \ldots, n-1\} \text{ and } \alpha_j,\ j \neq k,\ \text{such that} \quad p_k = \sum_{j\neq k} \alpha_j p_j.$$

Select $i \in \{0, \ldots, n-1\}$, $i \neq k$:

$$p_i^\mathsf{T} A p_k = \sum_{j\neq k} \alpha_j\, p_i^\mathsf{T} A p_j.$$

Since $\{p_0, \ldots, p_{n-1}\}$ are A-conjugate,

$$\Rightarrow \quad 0 = \alpha_i\, p_i^\mathsf{T} A p_i.$$

Since $A \succ 0$ and $p_i \neq 0 \Rightarrow p_i^\mathsf{T} A p_i > 0$

$$\Rightarrow\ \alpha_i = 0,\ i \neq k\ \Rightarrow\ p_k = 0 \quad (\Rightarrow\Leftarrow).$$

$\therefore$ $\{p_0, \ldots, p_{n-1}\}$ are linearly independent.

(b) α_k minimizer of $h(\alpha) = \varphi(x_k + \alpha p_k)$ means $h'(\alpha_k) = 0$.

- Computing $h'(\alpha) = \varphi'(x_k + \alpha p_k)^\mathsf{T} p_k$. We know $\varphi'(x) = Ax - b \Rightarrow \varphi'(x_k + \alpha p_k) = A(x_k + \alpha p_k) - b$.

$$\begin{aligned} \Rightarrow \; h'(\alpha) &= [A(x_k + \alpha p_k) - b]^\mathsf{T} p_k = p_k^\mathsf{T} A(x_k + \alpha p_k) - p_k^\mathsf{T} b \\ h'(\alpha) &= p_k^\mathsf{T} \underbrace{(Ax_k - b)}_{r_k} + \alpha\, p_k^\mathsf{T} A p_k \\ h'(\alpha_k) &= p_k^\mathsf{T} r_k + \Big(- \frac{r_k^\mathsf{T} p_k}{p_k^\mathsf{T} A p_k} \Big) p_k^\mathsf{T} A p_k \\ &= p_k^\mathsf{T} r_k - r_k^\mathsf{T} p_k = 0. \end{aligned}$$

$\therefore \alpha_k$ is a critical point of h.

But $h''(\alpha) = A \succ 0$.
$\therefore \alpha_k$ is the unique minimizer of h.

(c) Check notes.

P18 Practice *added*

Definitions.

- *Linear CG:* $r_0 = Ax_0 - b$, $p_0 = -r_0$ and, for $k \geq 0$, $\alpha_k = \dfrac{r_k^\mathsf{T} r_k}{p_k^\mathsf{T} A p_k}$, $x_{k+1} = x_k + \alpha_k p_k$, $r_{k+1} = r_k + \alpha_k A p_k$, $\beta_{k+1} = \dfrac{r_{k+1}^\mathsf{T} r_{k+1}}{r_k^\mathsf{T} r_k}$, $p_{k+1} = -r_{k+1} + \beta_{k+1} p_k$.

Problem. Linear CG by hand. Same data as E23: $A = \begin{pmatrix} 4 & 1 \\ 1 & 3 \end{pmatrix}$, $b = (1, 2)^\mathsf{T}$, $x_0 = (0, 0)^\mathsf{T}$.

a) Complete the second iteration: β_1, p_1, Ap_1, α_1, x_2, r_2.

b) Verify $r_1^\mathsf{T} p_0 = 0$, $p_1^\mathsf{T} A p_0 = 0$, $r_2 = 0$ and $x_2 = A^{-1} b$.

c) Steepest descent with exact step from the same x_0 takes the same first step. Explain which ingredient of CG makes the second step reach x^* (compare with V18 c).

(a)

$$x_0 = 0, \qquad r_0 = Ax_0 - b = -b = (-1, -2), \qquad p_0 = -r_0 = (1, 2).$$

Iteration 1.

$$r_0^{\mathsf{T}} r_0 = 1 + 4 = 5, \qquad Ap_0 = \begin{pmatrix} 4 & 1 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 6 \\ 7 \end{pmatrix}, \qquad p_0^{\mathsf{T}} Ap_0 = (1,2) \begin{pmatrix} 6 \\ 7 \end{pmatrix} = 20,$$

$$\alpha_0 = \frac{r_0^{\mathsf{T}} r_0}{p_0^{\mathsf{T}} Ap_0} = \frac{5}{20} = \frac{1}{4},$$

$$x_1 = x_0 + \alpha_0 p_0 = 0 + \left(\frac{1}{4}\right) \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1/4 \\ 1/2 \end{pmatrix},$$

$$r_1 = r_0 + \alpha_0 Ap_0 = (-1,-2) + \left(\frac{1}{4}\right) \begin{pmatrix} 6 \\ 7 \end{pmatrix} = \begin{pmatrix} -1+3/2 \\ -2+7/4 \end{pmatrix} = \begin{pmatrix} 1/2 \\ -1/4 \end{pmatrix},$$

$$r_0^{\mathsf{T}} r_1 = (-1,-2) \begin{pmatrix} 1/2 \\ -1/4 \end{pmatrix} = -\frac{1}{2} + \frac{1}{2} = 0,$$

$$r_1^{\mathsf{T}} r_1 = \frac{1}{4} + \frac{1}{16} = \frac{5}{16}, \qquad r_0^{\mathsf{T}} r_0 = 5, \qquad \beta_1 = \frac{r_1^{\mathsf{T}} r_1}{r_0^{\mathsf{T}} r_0} = \frac{5/16}{5} = \frac{1}{16},$$

$$p_1 = -r_1 + \beta_1 p_0 = \left(-\frac{1}{2}, \frac{1}{4}\right) + \frac{1}{16}(1,2) = \left(-\frac{1}{2} + \frac{1}{16}, \ \frac{1}{4} + \frac{1}{8}\right) = \left(-\frac{7}{16}, \ \frac{3}{8}\right),$$

$$p_1^{\mathsf{T}} Ap_0 = p_1^{\mathsf{T}} \begin{pmatrix} 4 & 1 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \left(-\frac{7}{16}, \frac{3}{8}\right) \begin{pmatrix} 6 \\ 7 \end{pmatrix} = -\frac{7 \cdot 6}{16} + \frac{3 \cdot 7}{8}$$

$$= -\frac{21}{8} + \frac{21}{8} = 0.$$

Iteration 2.

$$r_1^{\mathsf{T}} r_1 = \frac{5}{16}, \qquad Ap_1 = \begin{pmatrix} 4 & 1 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} -7/16 \\ 3/8 \end{pmatrix} = \begin{pmatrix} -7/4 + 3/8 \\ -7/16 + 9/8 \end{pmatrix} = \begin{pmatrix} -11/8 \\ 11/16 \end{pmatrix},$$

$$p_1^{\mathsf{T}} Ap_1 = \left(-\frac{7}{16}, \frac{3}{8}\right) \begin{pmatrix} -11/8 \\ 11/16 \end{pmatrix} = \frac{77}{16 \cdot 8} + \frac{33}{8 \cdot 16} = \frac{110}{16 \cdot 8} = \frac{55}{64},$$

$$\alpha_1 = \frac{r_1^{\mathsf{T}} r_1}{p_1^{\mathsf{T}} Ap_1} = \frac{5/16}{55/64} = \frac{4}{11},$$

$$x_2 = x_1 + \alpha_1 p_1 = \left(\frac{1}{4}, \frac{1}{2}\right) + \frac{4}{11}\left(-\frac{7}{16}, \frac{3}{8}\right) = \left(\frac{1}{4} - \frac{7}{44}, \ \frac{1}{2} + \frac{3}{22}\right) = \left(\frac{4}{44}, \ \frac{14}{22}\right),$$

$$x_2 = \left(\frac{1}{11}, \ \frac{7}{11}\right).$$

- $Ax_2 - b = \begin{pmatrix} 4 & 1 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 1/11 \\ 7/11 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \left(\frac{11}{11} - 1, \ \frac{22}{11} - 2\right) = 0 \quad \therefore x_2 = x^*.$

$$r_2 = Ax_2 - b = 0 = r_1 + \alpha_1 Ap_1 = \left(\frac{1}{2}, -\frac{1}{4}\right) + \frac{4}{11}\left(-\frac{11}{8}, \ \frac{11}{16}\right) = \left(\frac{1}{2}, -\frac{1}{4}\right) + \left(-\frac{1}{2}, \ \frac{1}{4}\right) = 0,$$

$$\beta_2 = \frac{r_2^{\mathsf{T}} r_2}{r_1^{\mathsf{T}} r_1} = 0 \ \Rightarrow \ p_2 = -r_2 + \beta_2 p_1 = -0 + 0\, p_1 = 0.$$

```
(env) [glozanoac@bell src]$ python3 linear_cg.py
[k=10] r=[ 0.500 -0.250], p=[-0.438  0.375], beta=     0.062
[k=9] r=[0.000 0.000], p=[-0.000  0.000], beta=     0.000
[0.091 0.636] 1
(env) [glozanoac@bell src]$
```

Algorithm implemented in **linear_cg.py** and **P18.py**

T50 PROOF *ch6, Lemma 2.1, p. 3*

Setting. Let $\varphi(x) = \frac{1}{2}x^\mathsf{T}Ax - b^\mathsf{T}x$ with $A \in \mathbb{R}^{n\times n}$ symmetric positive definite, $x^* = A^{-1}b$, and residual $r_k = Ax_k - b = \nabla\varphi(x_k)$. $x_k \in \mathbb{R}^n$, $p_k \neq 0$, α_k the minimizer of $\alpha \mapsto \varphi(x_k + \alpha p_k)$, and $x_{k+1} = x_k + \alpha_k p_k$.
Problem.

a) Prove that $\alpha_k = -\dfrac{r_k^\mathsf{T}p_k}{p_k^\mathsf{T}Ap_k}$.

b) Prove that $r_{k+1} = r_k + \alpha_k Ap_k$ and $r_{k+1}^\mathsf{T}p_k = 0$.

(a) Done in T49 (b).

(b)

- **Affirmation:** $r_{k+1} = r_k + \alpha_k Ap_k$.

$$\begin{aligned} x_{k+1} &= x_k + \alpha_k p_k \\ Ax_{k+1} &= Ax_k + \alpha_k Ap_k \\ Ax_{k+1} - b &= (Ax_k - b) + \alpha_k Ap_k \\ r_{k+1} &= r_k + \alpha_k Ap_k. \end{aligned}$$

- **Affirmation:** $r_{k+1}^\mathsf{T}p_k = 0$.

$$\begin{aligned} r_{k+1} &= r_k + \alpha_k Ap_k \\ p_k^\mathsf{T}r_{k+1} &= p_k^\mathsf{T}r_k + \alpha_k\, p_k^\mathsf{T}Ap_k \\ &= p_k^\mathsf{T}r_k + \Big(-\frac{r_k^\mathsf{T}p_k}{p_k^\mathsf{T}Ap_k}\Big)p_k^\mathsf{T}Ap_k \\ &= p_k^\mathsf{T}r_k - r_k^\mathsf{T}p_k. \end{aligned}$$

But $\mathbb{R} \ni p_k^\mathsf{T}r_k = (p_k^\mathsf{T}r_k)^\mathsf{T} = r_k^\mathsf{T}p_k$

$$\Rightarrow\ p_k^\mathsf{T}r_{k+1} = 0.$$

T51 Proof *ch6, Lemma 2.3, pp. 5–6*

Setting. Let $\varphi(x) = \frac{1}{2}x^\mathsf{T}Ax - b^\mathsf{T}x$ with $A \in \mathbb{R}^{n\times n}$ symmetric positive definite, $x^* = A^{-1}b$, and residual $r_k = Ax_k - b = \nabla\varphi(x_k)$.
Definitions.

- *Linear CG:* $r_0 = Ax_0 - b$, $p_0 = -r_0$ and, for $k \geq 0$, $\alpha_k = \dfrac{r_k^\mathsf{T}r_k}{p_k^\mathsf{T}Ap_k}$, $x_{k+1} = x_k + \alpha_k p_k$, $r_{k+1} = r_k + \alpha_k Ap_k$, $\beta_{k+1} = \dfrac{r_{k+1}^\mathsf{T}r_{k+1}}{r_k^\mathsf{T}r_k}$, $p_{k+1} = -r_{k+1} + \beta_{k+1}p_k$.

Problem. Assume $r_0, r_1 \neq 0$. Prove that $p_0^\mathsf{T}Ap_1 = 0$.

Note. The notes write $r_0^\mathsf{T}Ap_0 = \frac{r_0^\mathsf{T}r_0}{\alpha_0}$; the correct value is $r_0^\mathsf{T}Ap_0 = \frac{r_0^\mathsf{T}r_1 - r_0^\mathsf{T}r_0}{\alpha_0} = -\frac{r_0^\mathsf{T}r_0}{\alpha_0}$. With the right sign, $p_0^\mathsf{T}Ap_0 = -r_0^\mathsf{T}Ap_0 = \frac{\|r_0\|^2}{\alpha_0}$ and $p_0^\mathsf{T}Ap_1 = -\frac{\|r_1\|^2}{\alpha_0} + \beta_1\frac{\|r_0\|^2}{\alpha_0} = 0$.

$$\begin{aligned} p_0^\mathsf{T}Ap_1 &= p_0^\mathsf{T}A(-r_1 + \beta_1 p_0) \\ &= -p_0^\mathsf{T}Ar_1 + \beta_1\, p_0^\mathsf{T}Ap_0 \\ &= -r_1^\mathsf{T}Ap_0 + \beta_1\, p_0^\mathsf{T}Ap_0. \end{aligned}$$

$$r_1 = r_0 + \alpha_0 A p_0, \qquad A p_0 = \frac{r_1 - r_0}{\alpha_0}$$

$$\begin{aligned} p_0^\mathsf{T} A p_1 &= -r_1^\mathsf{T}\Big(\frac{r_1 - r_0}{\alpha_0}\Big) + \beta_1 (-r_0)^\mathsf{T}\Big(\frac{r_1 - r_0}{\alpha_0}\Big) \\ &= -\frac{r_1^\mathsf{T} r_1 + r_1^\mathsf{T} r_0}{\alpha_0} + \frac{\beta_1}{\alpha_0}(-r_0^\mathsf{T} r_1 + r_0^\mathsf{T} r_0). \end{aligned}$$

Affirmation: $r_1^\mathsf{T} r_0 = 0.$

$$\begin{aligned} r_1^\mathsf{T} r_0 = r_0^\mathsf{T} r_1 &= r_0^\mathsf{T}(r_0 + \alpha_0 A p_0) \\ &= r_0^\mathsf{T} r_0 + \alpha_0 (-p_0)^\mathsf{T} A p_0 \\ &= r_0^\mathsf{T} r_0 + \Big(\frac{r_0^\mathsf{T} r_0}{p_0^\mathsf{T} A p_0}\Big)(-p_0^\mathsf{T} A p_0) \\ &= r_0^\mathsf{T} r_0 - r_0^\mathsf{T} r_0 = 0. \end{aligned}$$

$$\Rightarrow \; p_0^\mathsf{T} A p_1 = -\frac{r_1^\mathsf{T} r_1}{\alpha_0} + \frac{\beta_1}{\alpha_0}(+ r_0^\mathsf{T} r_0)$$

$$\alpha_0 = \frac{r_0^\mathsf{T} r_0}{p_0^\mathsf{T} A p_0}, \qquad \beta_1 = \frac{r_1^\mathsf{T} r_1}{r_0^\mathsf{T} r_0}$$

$$\begin{aligned} p_0^\mathsf{T} A p_1 &= -\frac{r_1^\mathsf{T} r_1}{\alpha_0} + \Big(\frac{r_1^\mathsf{T} r_1}{r_0^\mathsf{T} r_0}\Big)\Big(\frac{r_0^\mathsf{T} r_0}{\alpha_0}\Big) \\ &= -\frac{r_1^\mathsf{T} r_1}{\alpha_0} + \frac{r_1^\mathsf{T} r_1}{\alpha_0} = 0. \end{aligned}$$

$\therefore$ p_0 and p_1 are A-conjugate.